



INTERNSHIP PROJECT

Afforestation and Deforestation Effects on European Topsoil Organic Carbon

Linking LUCAS soil data with a 1950s-onward land-cover reconstruction across a European climate gradient

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Johann Heinrich von Thünen Institute · Bundesallee 68 | Project duration: 15 March 2026 – 15 June 2026

The question: does land-use change alter soil carbon?

Soil organic carbon (SOC) is the largest terrestrial carbon store. Converting land to or from forest may change how much carbon the topsoil holds — but the effect is slow, plays out over decades, and is tangled up with climate.

- 1 After afforestation or deforestation, how do long-term SOC levels in mineral topsoil differ from persistent land use?
- 2 How does SOC vary along a time-since-transition gradient?
- 3 How is this influenced by climate — mean annual temperature and precipitation?

Study area: Spain, France, Baden-Württemberg (Germany) and Sweden — a gradient from warm-dry south to cool-wet north.

Two datasets, linked by site

LUCAS soil

Mineral topsoil organic carbon (SOC), its trend over 2009-2018 (oc_slope), and climate: mean annual temperature & precipitation. ~900 sites.

My land-use reconstruction

Land cover at each site from the 1950s to present (3+ time points), built from historical aerial imagery — the true long-term change.

Why both

LUCAS's own change record covers only ~2006-2018; my reconstruction captures change going back decades, which LUCAS misses.

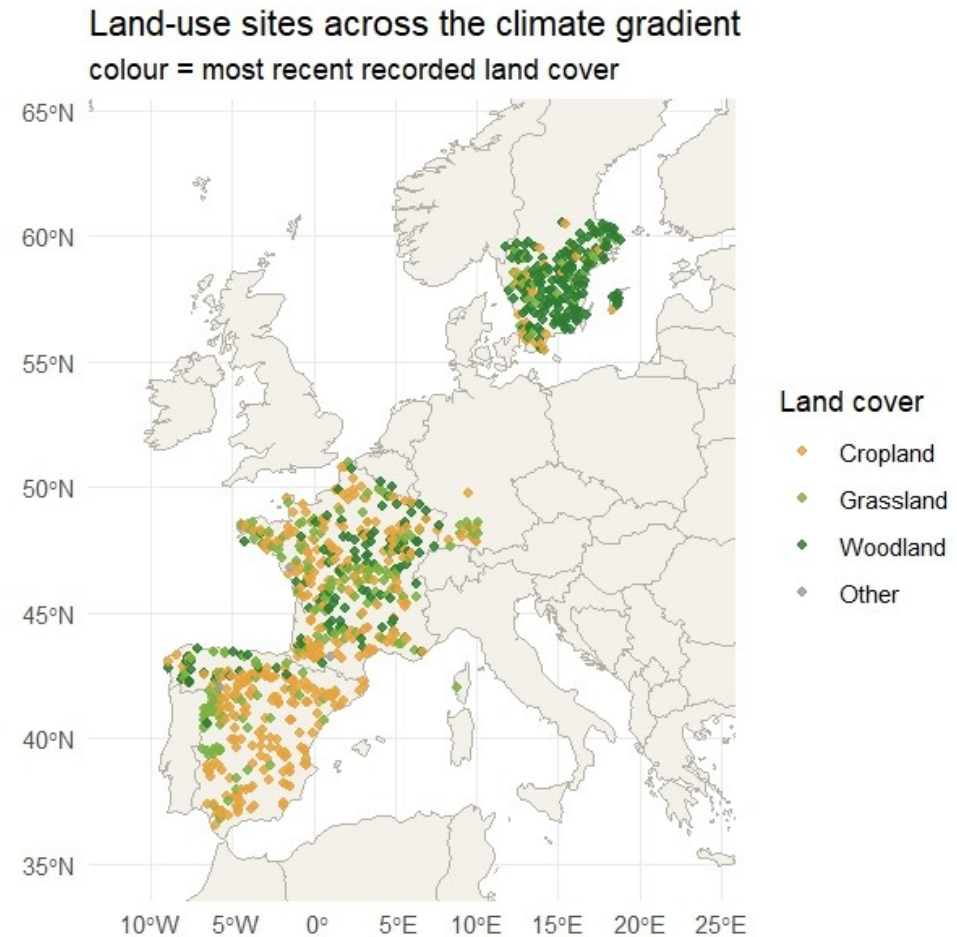
Linking

Merged on the shared site ID: SOC + climate (LUCAS) beside land-use history (my file). Analysis in R.

Sites span a south-to-north climate gradient

- Each point is a LUCAS site, coloured by its current land cover.
- Four regions: Spain and France (south/west), Baden-Württemberg, and Sweden (north).
- This spread is what lets us separate the land-use effect from the climate effect.

Takeaway A wide climate gradient with all three land-cover types is the foundation of the analysis.

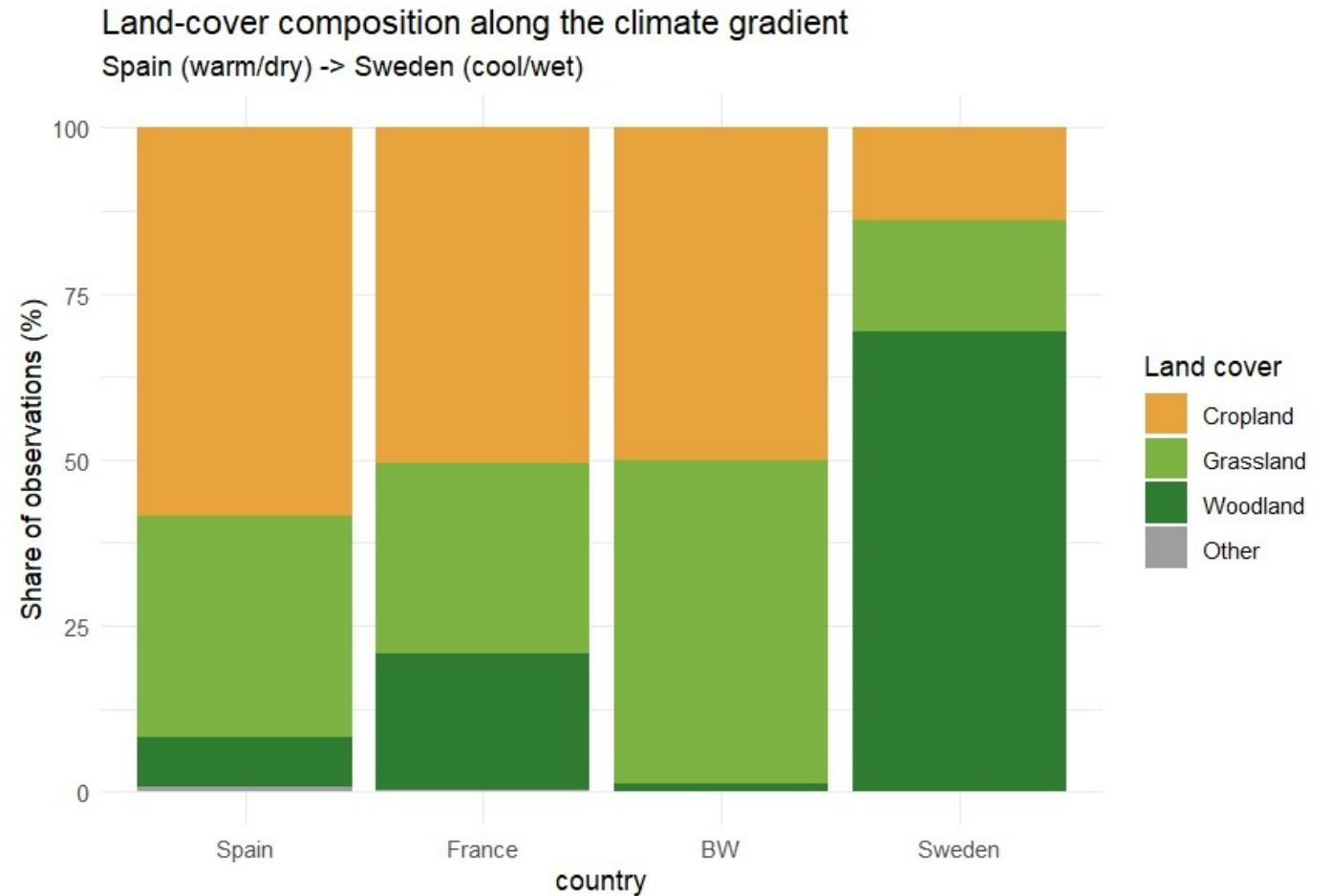


THE DATA

Land-cover composition differs by country

- Sweden is forest-dominated; Spain and France are more cropland and grassland.
- Baden-Württemberg is a smaller, mixed set of sites.
- These differences matter — they shape where each type of land-use change can occur.

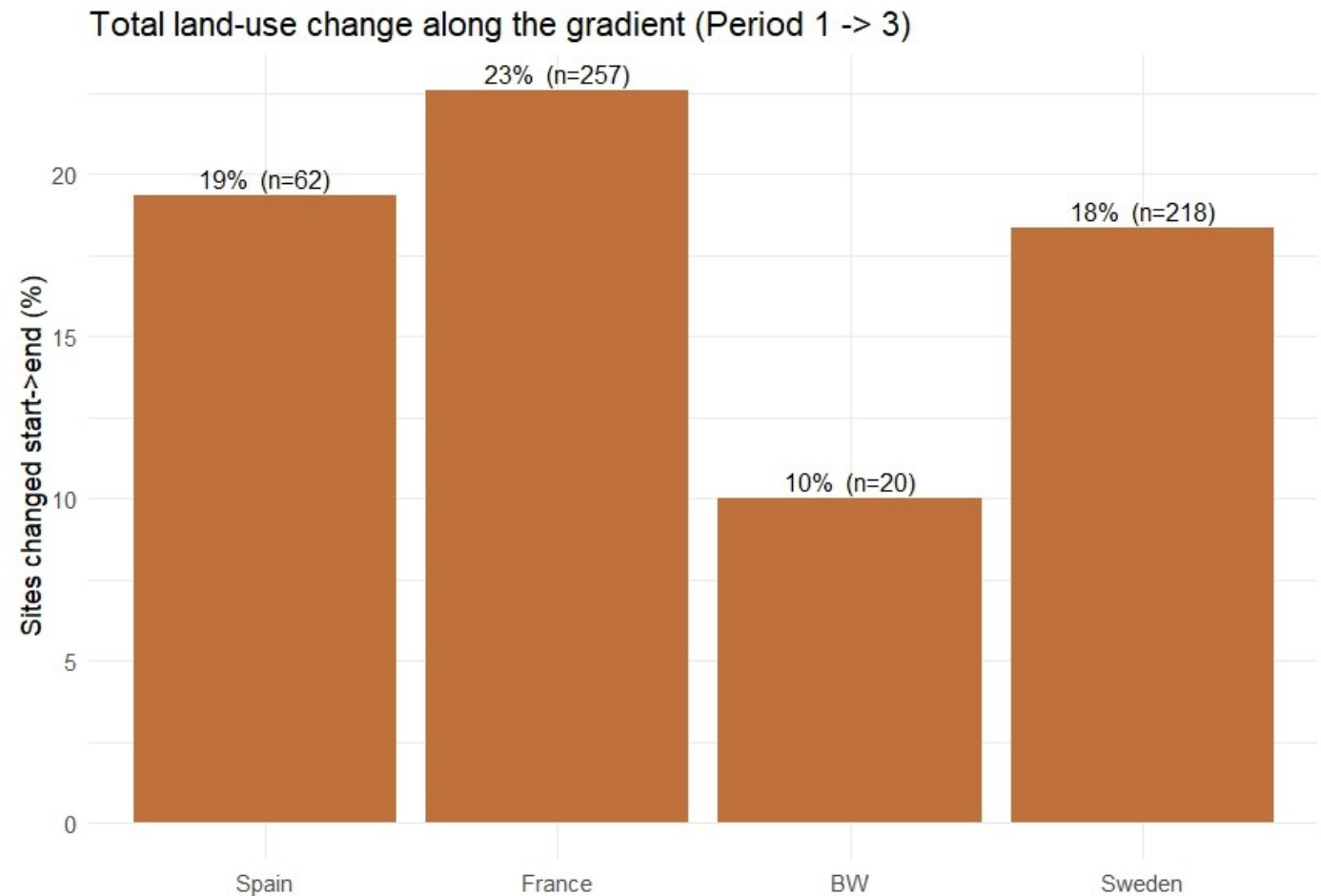
Takeaway Each country contributes a different land-cover mix to the gradient.



Land use has shifted since the 1950s

- From the reconstruction: most sites are persistent, but a meaningful minority changed.
- Afforestation (→ woodland) and deforestation (woodland →) are the changes of interest.
- Crop-grass swaps also occur, mostly in the agricultural south.

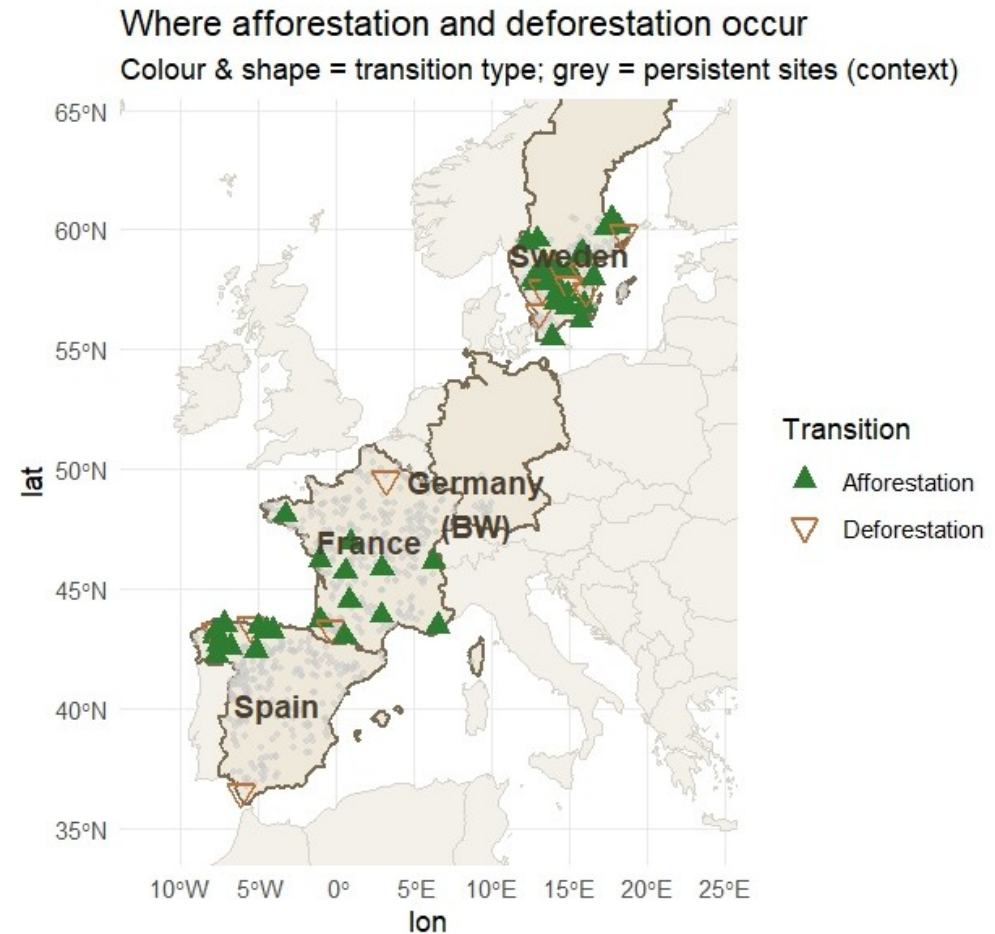
Takeaway Real, decades-long land-use change is captured — the basis for the SOC comparison.



Where afforestation and deforestation occur

- Colour & shape = transition type; country borders and names mark the regions.
- Afforestation clusters in Sweden and northern Spain; deforestation is rare everywhere.
- Land-use change is regional, not evenly spread across the gradient.

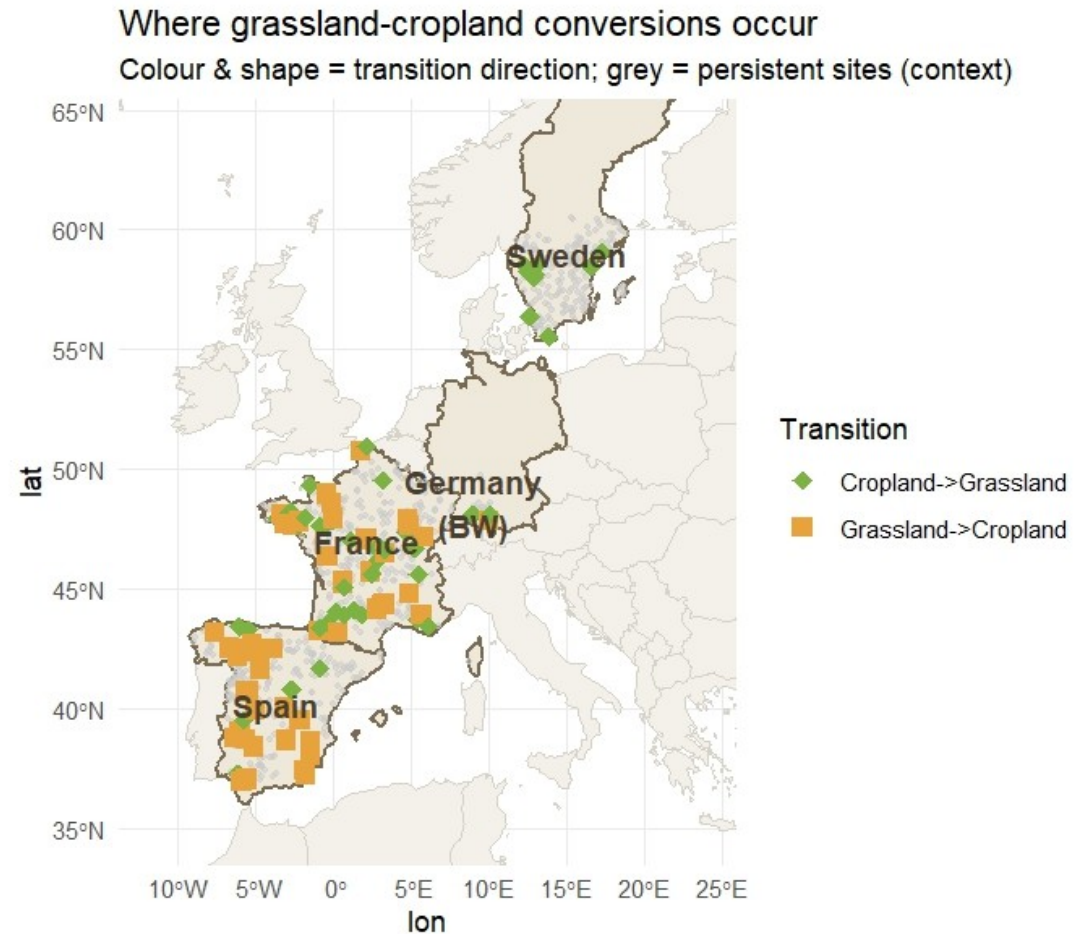
Takeaway Afforestation is concentrated in the north and in Spain; deforestation is uncommon.



Where grassland-cropland conversions occur

- The agricultural swaps (square = grass→crop, diamond = crop→grass).
- These cluster in France and Spain — the agricultural heart of the dataset.
- Sweden shows almost none: a southern / central-European pattern.

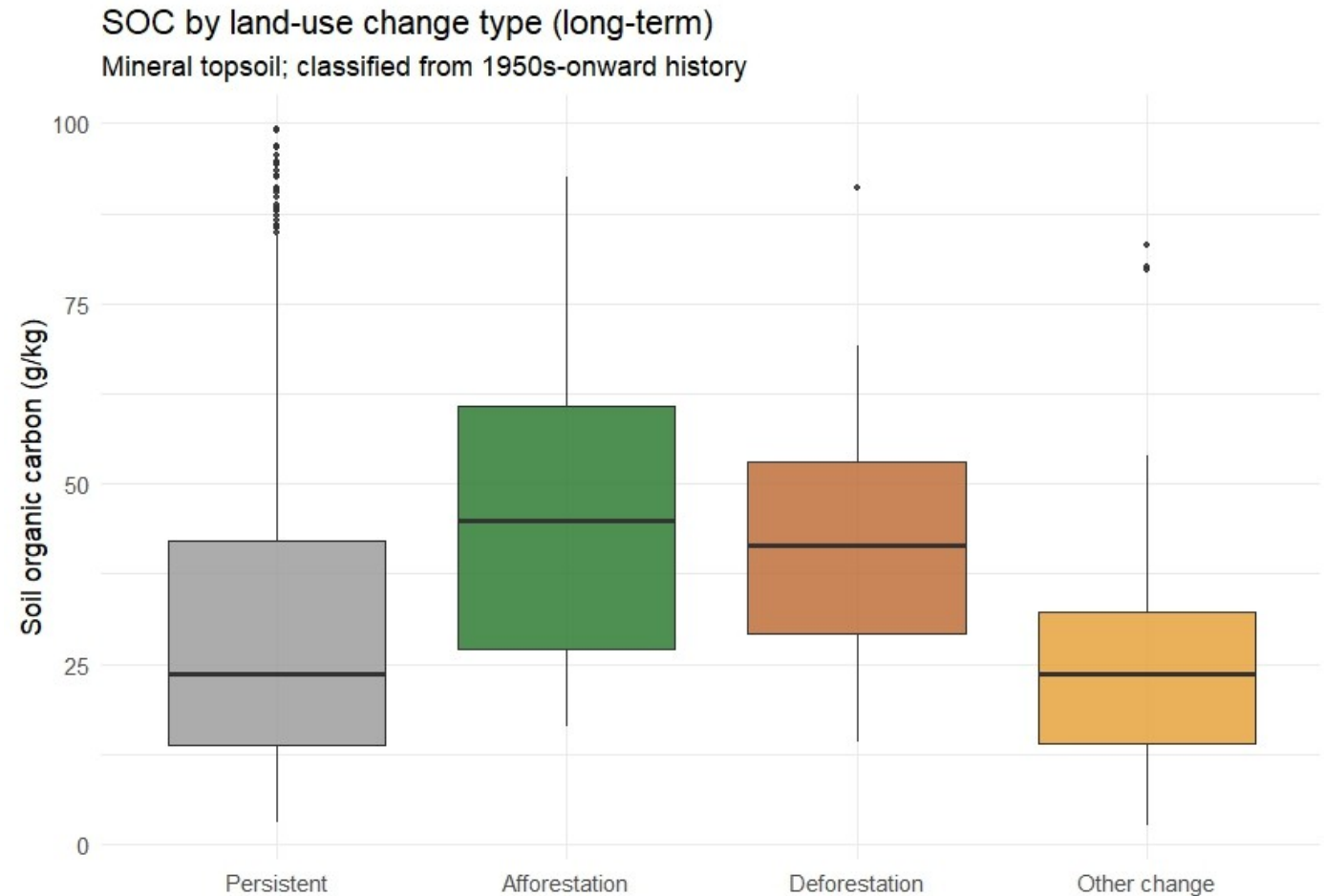
Takeaway Crop-grass conversion is a French and Spanish phenomenon, absent in the north.



SOC is higher where woodland is involved

- Afforestation and deforestation sites hold roughly double the SOC of persistent crop/grass sites.
- The woodland signal dominates: forest-linked soils are carbon-rich.
- Sample sizes: persistent n=737, afforestation n=47, deforestation n=12.

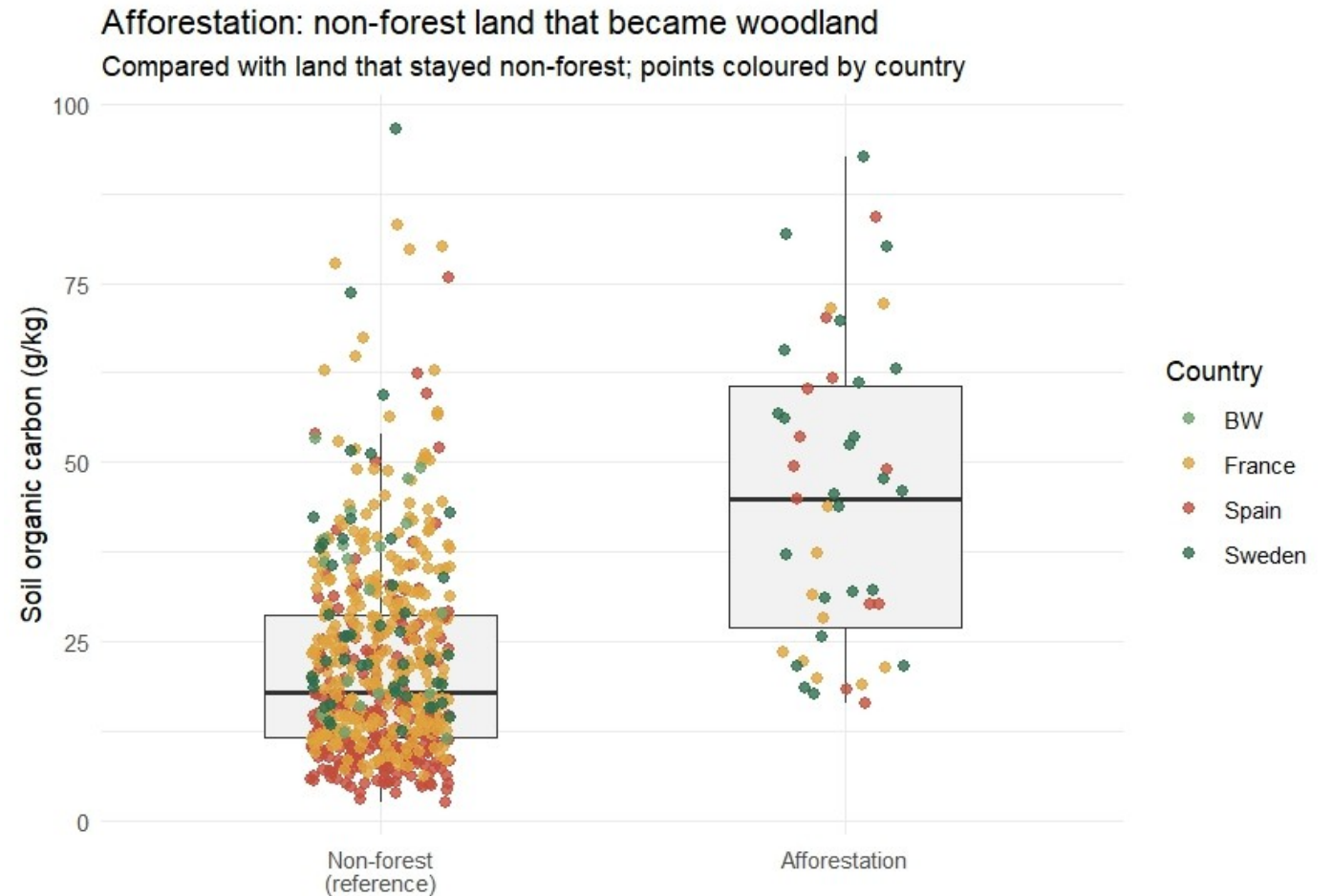
Takeaway Land-use change involving woodland leaves a lasting, high-carbon soil signature.



Afforested land holds more carbon than non-forest

- Afforestation sites compared with land that stayed non-forest (their starting point).
- Afforested sites sit clearly higher (~45 vs ~18 g/kg) - decades of forest build carbon.
- Points coloured by country: Sweden, Spain and France make up the 47 sites.

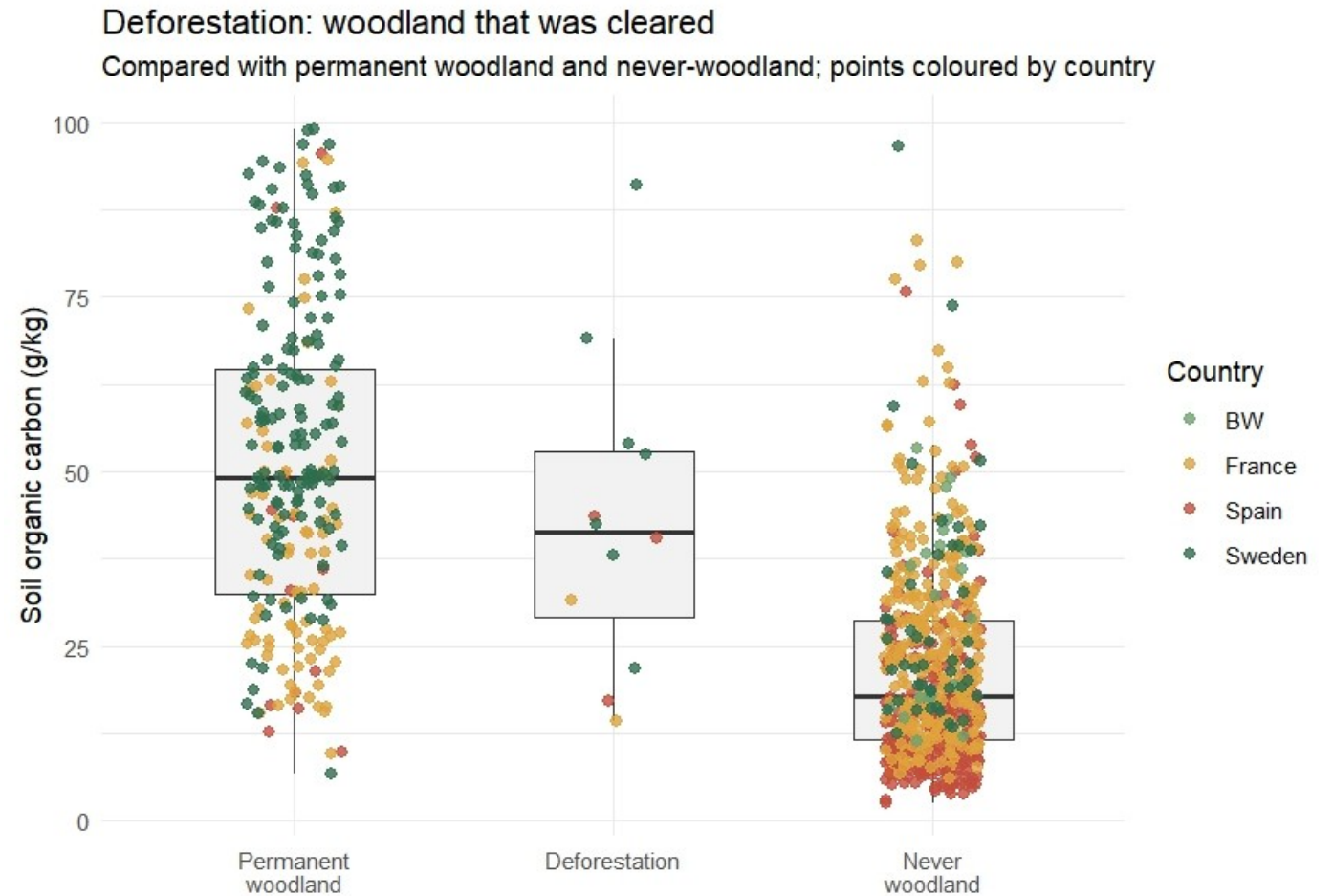
Takeaway Converting land to forest is associated with markedly higher topsoil carbon.



Cleared woodland sits between forest and non-forest

- Deforestation vs permanent woodland (kept forest) and never-woodland.
- Deforested sites sit below permanent woodland but above never-woodland.
- Only n=12 deforested sites — indicative, interpret with caution.

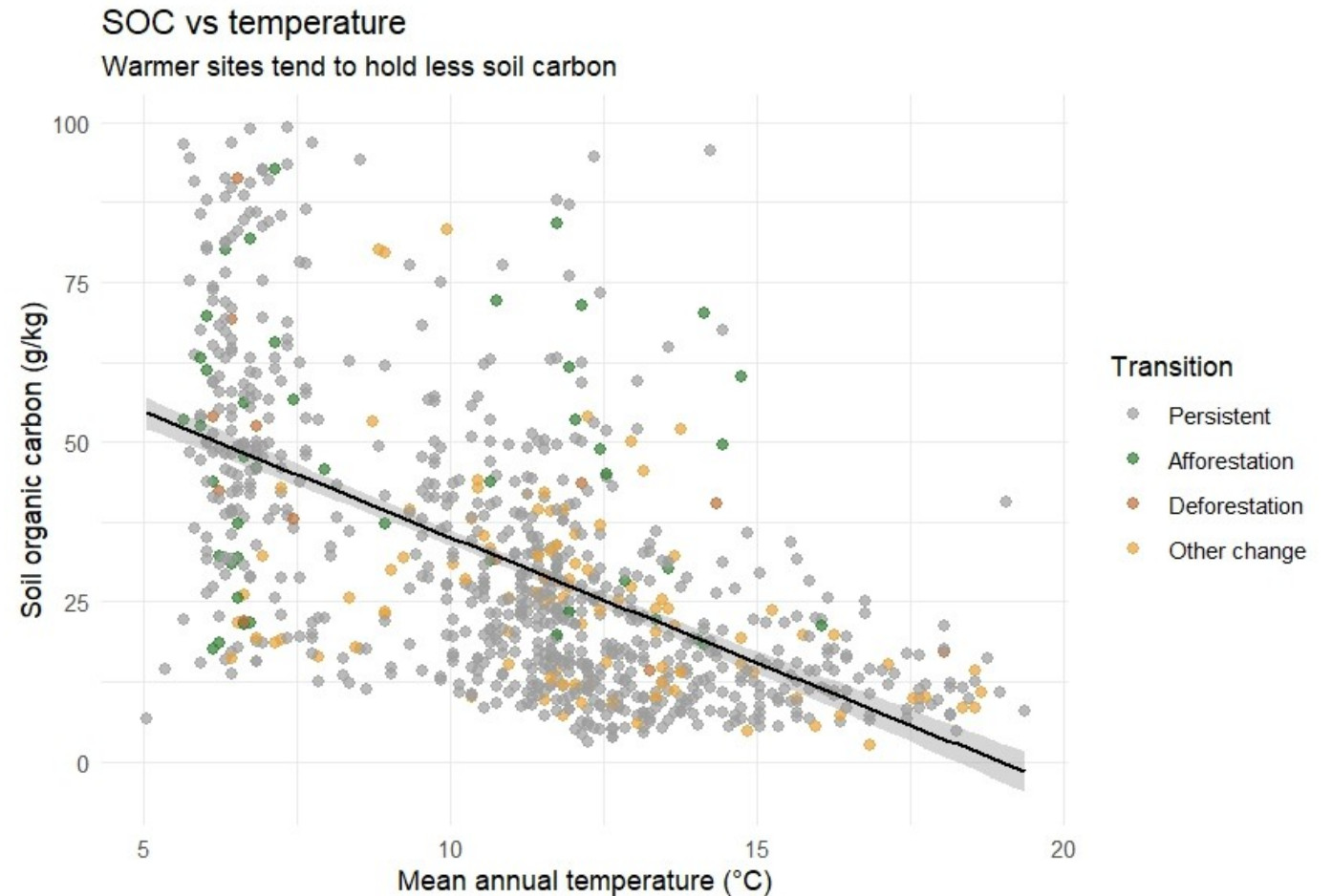
Takeaway Clearing forest is linked to carbon loss, but the small sample makes it suggestive.



Temperature is the strongest control on SOC

- Clear negative relationship ($r = -0.58$): warmer sites hold far less carbon.
- Warmth speeds decomposition, so carbon does not accumulate.
- Precipitation adds a weaker positive effect ($r = 0.24$).

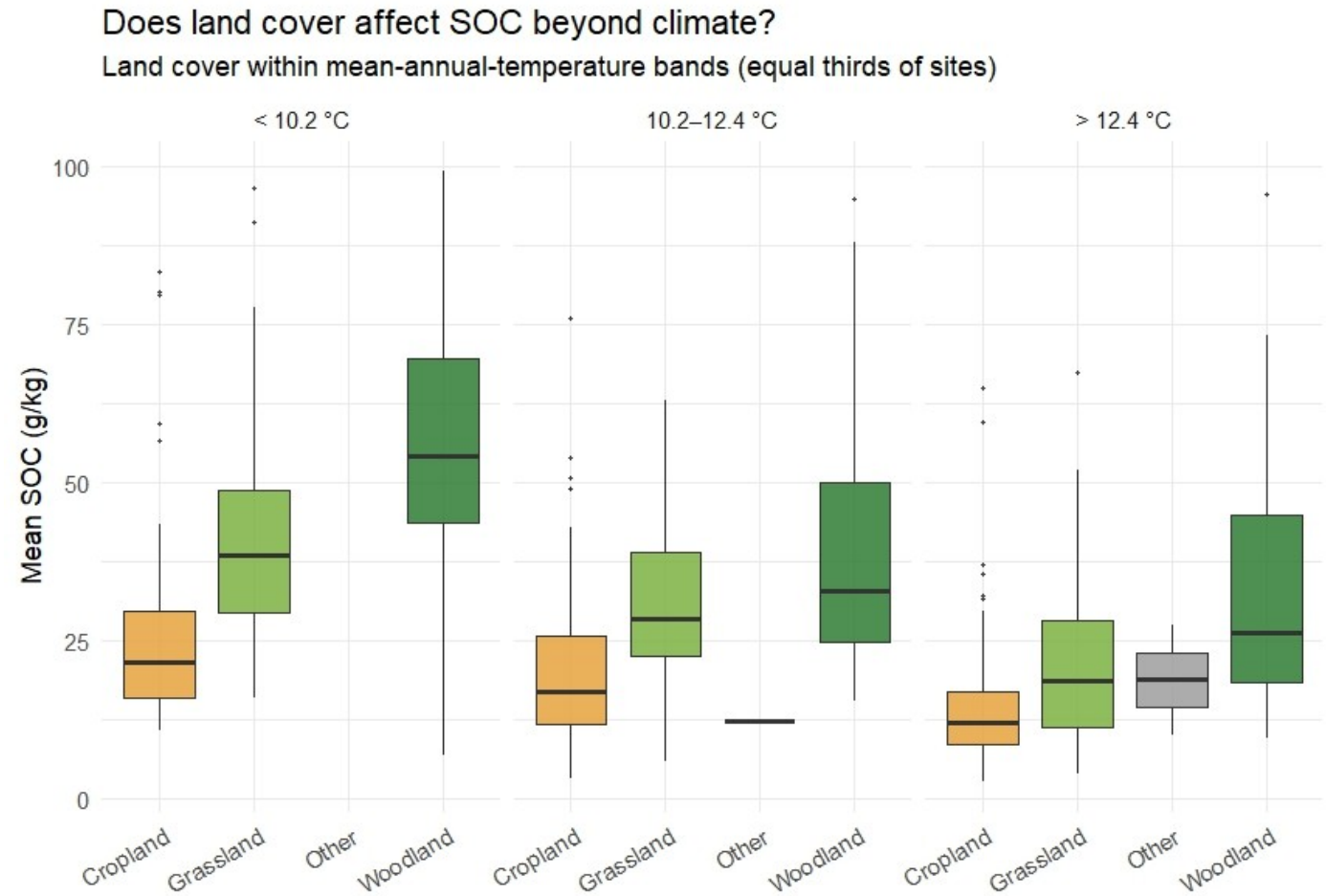
Takeaway Climate sets the baseline - warm sites are carbon-poor, cool sites carbon-rich.



Land cover matters on top of climate

- SOC by land cover, repeated within real temperature bands (< 10.2, 10.2-12.4, > 12.4 °C).
- In every band the order holds: woodland highest, cropland lowest.
- All boxes drop as temperature rises - the climate effect runs underneath.

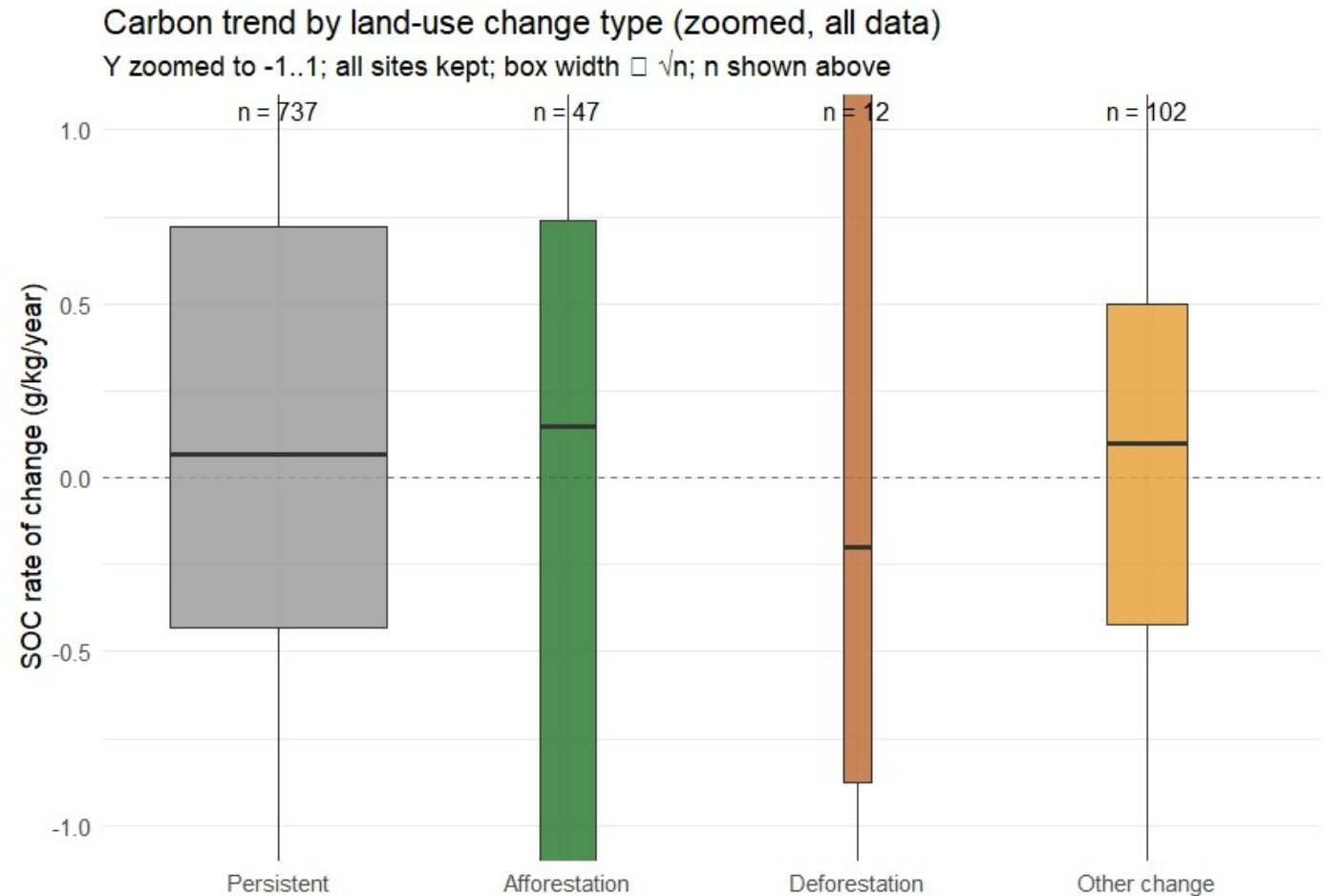
Takeaway Land cover affects carbon independently of climate - the ranking holds in every band.



Is carbon currently rising or falling?

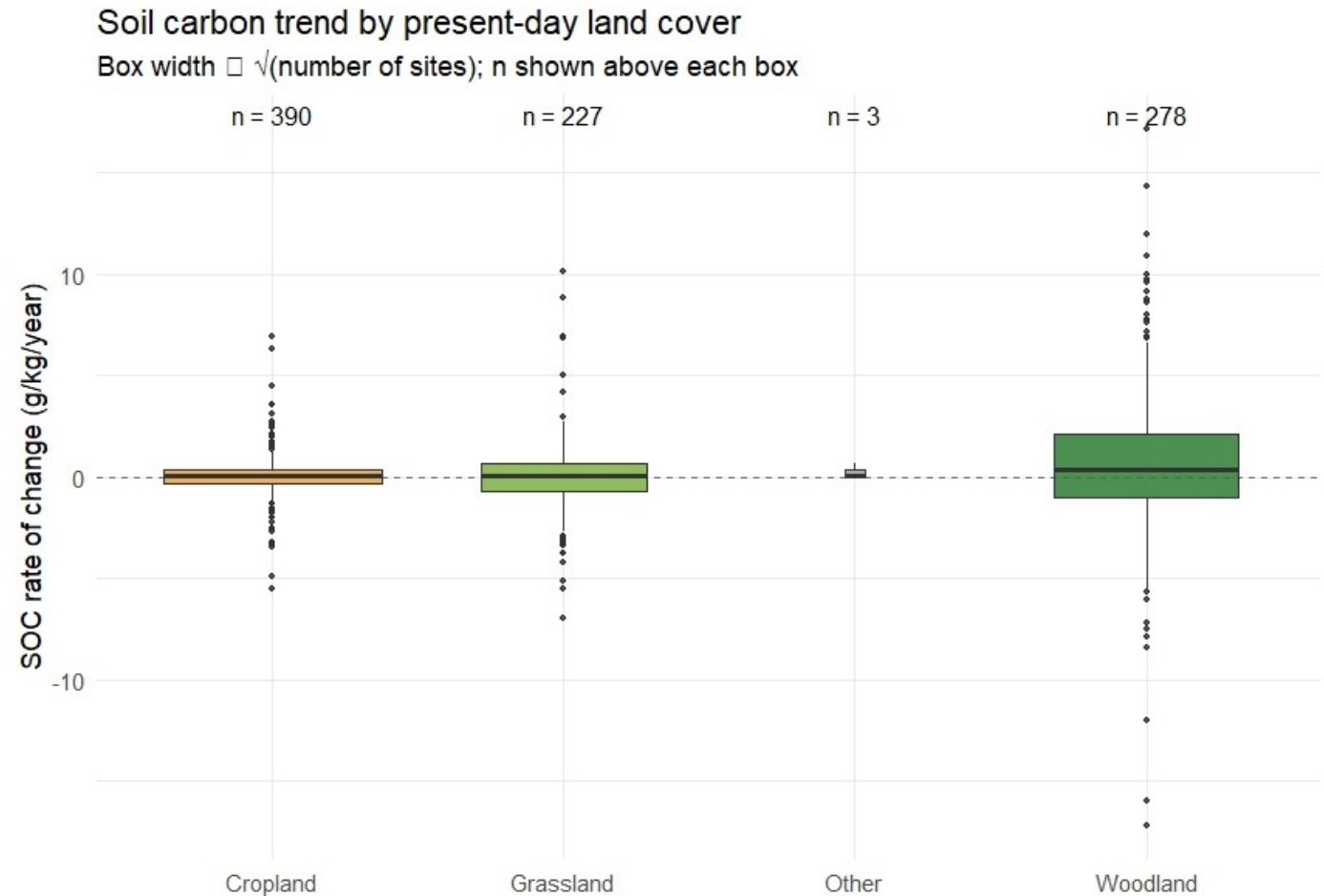
- A new outcome: oc_slope = the recent rate of carbon change (2009-2018).
- Implausible slopes removed as sampling error; box width $\propto \sqrt{n}$, n shown above.
- By change category, all groups sit near zero - past change does not predict the recent trend.

Takeaway Long-term change category does not clearly predict whether carbon is rising now.



Present-day woodland is still gaining carbon

- Same trend, but by what the land is TODAY rather than how it changed.
- Woodland sits above zero - forest soils are actively gaining carbon.
- Cropland and grassland are roughly steady.

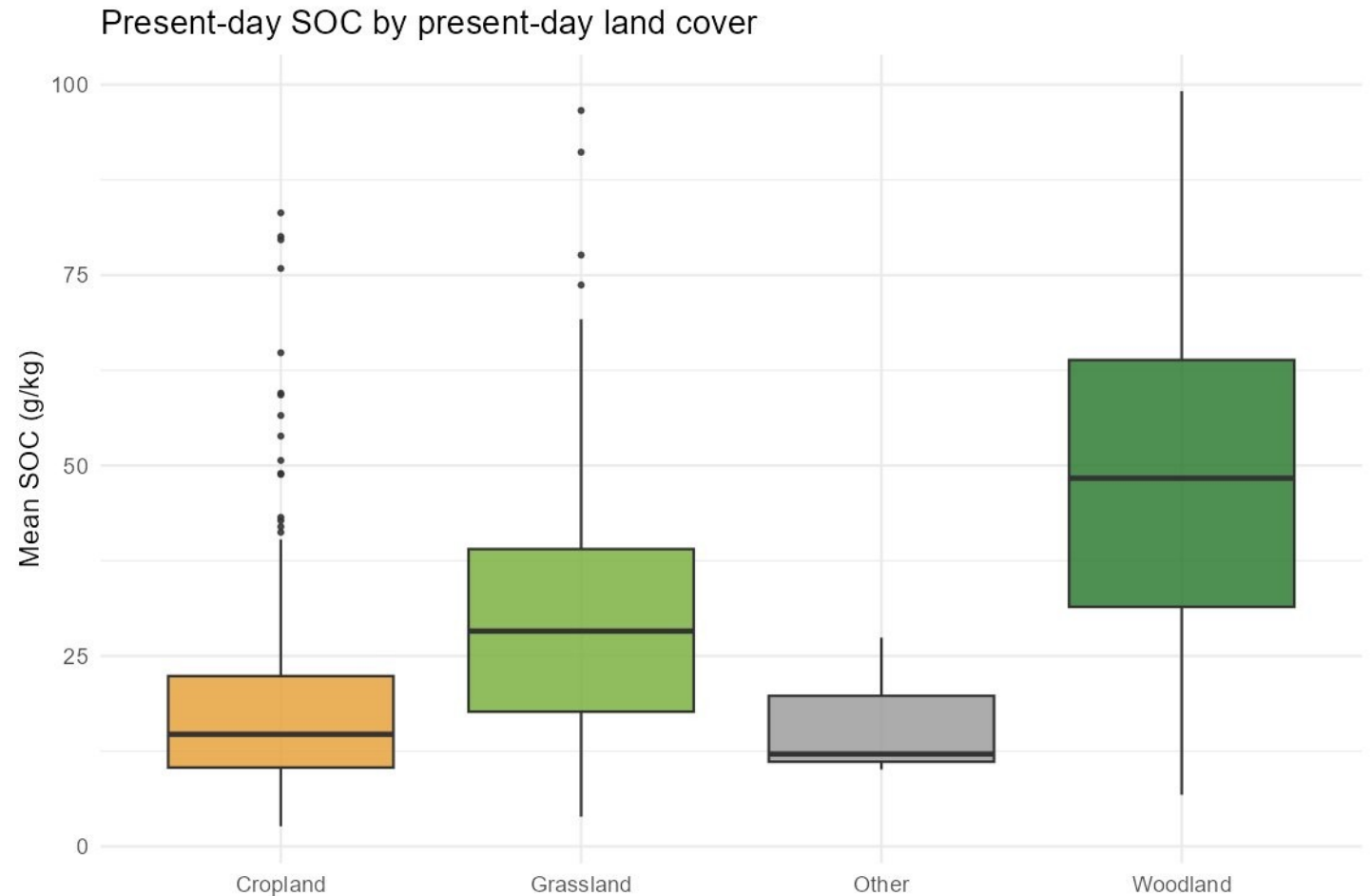


Takeaway What the land is today predicts the carbon trend better than its history does.

Woodland holds ~3x the carbon of cropland

- Present-day SOC level by present-day land cover.
- Woodland median ~48 g/kg, grassland ~28, cropland ~15.
- The clearest single result: forest >> grassland > cropland.

Takeaway Today's land cover is a strong, direct predictor of how much carbon the soil holds.



WHAT THE RESEARCH SHOWS

1 Woodland soils are carbon-rich

Afforestation and deforestation sites hold ~2x the SOC of crop/grass land - the forest signature persists for decades.

2 Afforestation builds carbon

Afforested land holds far more carbon than the non-forest it came from; deforestation shows the reverse (small sample).

3 Climate sets the baseline

SOC falls strongly with temperature ($r = -0.58$); but land cover matters independently in every climate band.

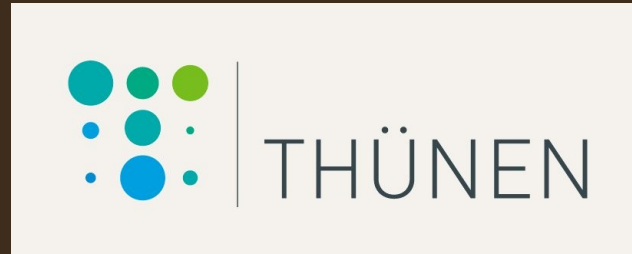
4 Change is regional

Afforestation clusters in Sweden and Spain; crop-grass swaps in France and Spain - the type of change varies geographically.

5 Trend follows the present

Woodland is still gaining carbon today; the recent trend tracks present-day cover, not historical change.

Caveats: deforestation $n=12$ (indicative only); SOC trend covers 2009-2018 only; Spain's record ends ~2000; relationships are correlational, not causal.



Thank you

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